

Disha Nikam

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EDUCATION

University of Pennsylvania | GPA: 3.70/4.0

Philadelphia, US

M.S.E. in Computer and Information Science

Aug 2025 – May 2027

- Graduate Research Assistant: Han AI Lab at Penn Medicine (Jan 2025-Present)
- Coursework: Trustworthy Machine Learning, Artificial Intelligence, Applied Machine Learning, Big Data Analytics

PES University | GPA: 9.28/10

India

B.Tech in Computer Science

Dec 2020 – May 2024

- Honors: Professor MRD Scholarship awarded 6 semesters in a row; Among top 20% of the batch
- Activities: Director of Editorial Service (Rotaract Club), IEEE CS Club

PUBLICATIONS

BGKnow-Medical Chatbot: A Hybrid Approach based on Knowledge Graph and GPT-2

ADCIS 2023

- Proceedings published in Scopus-Indexed Springer Series [Link]

WORK EXPERIENCE

Technology Consultant | PwC India, Bangalore

Sept 2024 – Apr 2025

- Architected and delivered speech-enabled, multilingual RAG chatbots in Python supporting 10+ languages, serving 500K+ citizens as part of a Government of India initiative and cutting client query turnaround time by 40%
- Strengthened AI governance by implementing LLM evaluation frameworks and guardrails including hallucination detection and toxicity filters, ensuring compliant and reliable deployment across enterprise use cases
- Containerized AI solutions with Docker and deployed on Microsoft Azure, reducing deployment time by 60% and improving system scalability for high-concurrency enterprise workloads

Software Developer Intern | Walmart Global Tech, Bangalore

Jan 2024 – June 2024

- Built and deployed a production RAG-based chatbot with React.js frontend, reducing average knowledge retrieval time by 35% and boosting employee productivity across 40+ users
- Applied RAGAS evaluation metrics, progressively improving context-aware response accuracy and retrieval precision over baseline

Summer Intern | Big Basket, Bangalore

June 2022 – Aug 2022

- Analyzed sales data across 12+ product categories using Python and SQL, uncovering trends that directly informed pricing and marketing strategies
- Delivered actionable recommendations adopted by the business team, contributing to measurable improvement in sales performance and reduced losses in 3+ underperforming segments

Software Developer Intern | G-KnowMe, Bangalore

June 2021 – Aug 2021

- Built a Flask web application that automated PubMed mining across 1,000+ clinical trial records, reducing manual data extraction effort by 70%
- Improved biomedical research efficiency by developing an NLP-based tagging pipeline for structured article retrieval and exploration

PROJECTS

Agent Evaluation Framework: Multi-Agent Benchmarking & LLM-as-Judge

Feb 2026 – March 2026

- Built an end-to-end **agent evaluation framework** for multi-domain assistants (fitness, nutrition, productivity, health, general), with a **30-prompt** benchmark, YAML-configured personas, and a **5-dimensional** rubric (accuracy, helpfulness, safety, personalization, latency) via LLM-as-judge
- Engineered a **reliability pipeline** across **3** repeated runs with variance and per-agent consistency metrics, combining lexical/BERTScore-style checks with optional **LLM functional-equivalence** judging; outputs structured CSV/JSON for engineering and golden-answer generation
- Shipped a **Streamlit** dashboard for interactive exploration of scores, reliability, and golden answers, plus automated workflows to generate **golden responses** and engineering notes for weak or failing prompts

Adaptive Learning Tutor: Claude Agent with Tools & Adaptive Curriculum

Jan 2026 – Feb 2026

- Built a **Claude-based tutor agent** (Anthropic API) with **tool use** and **multi-turn** memory—not one-shot chat—driving onboarding, diagnostic questions, **personalized multi-week curricula**, and session-by-session teaching with MCQ and open-ended quizzes
- Implemented **adaptive learning logic** that tracks weak areas and knowledge gaps via persisted progress, adjusts pacing from quiz outcomes, and optionally augments modules with **Tavily** web search for up-to-date resources; includes **demo mode** when the API is unavailable
- Shipped a full-stack app: **FastAPI** backend (agent loop, tools, JSON-backed user storage) and **React 19 + TypeScript + Vite** frontend with onboarding, session flow, and a **progress dashboard** (curriculum view, completion, time spent, areas to revisit)

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, Java, JavaScript, HTML/CSS, React Native, Linux

AI/ML & Data Science: TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, Pandas, NumPy

NLP & Computer Vision: NLTK, Matplotlib, SpaCy, SciSpacy, OpenCV, Pytesseract

Tools & Databases: Cursor, Codex, ClaudeCode, Docker, Flask, Streamlit, MATLAB, MySQL, Neo4j, py2neo, Git, Slurm